

# PROBLEMAS 3.

1.

Calcular  $T_{ex}$  y Speedup

$$T_{ex} = I \times CPI \times T_c$$

$$T_{c \text{ multi}} = 68 \text{ ps}$$

$$T_{c \text{ mono}} = 131 \text{ ps}$$

$$i) T_{ex \text{ Multi}} = 8.5 \cdot 68 \text{ ps} = 2720 \text{ ps} \rightarrow 0.367 \text{ GHz}$$

$$T_{ex \text{ mono}} = 8 \cdot 1 \cdot 131 = 1048 \text{ ps} \rightarrow 0.954 \text{ GHz}$$

$$ii) T_{ex \text{ multi}} = 5 \cdot 2 \cdot 68 = 680 \text{ ps} \rightarrow 1.47 \text{ GHz}$$

$$T_{ex \text{ mono}} = 5 \cdot 1 \cdot 131 = 655 \text{ ps} \rightarrow 1.526 \text{ GHz}$$

$$iii) T_{ex \text{ multi}} = 4 \cdot 3 \cdot 68 = 816 \text{ ps} \rightarrow 1.225 \text{ GHz}$$

$$T_{ex \text{ mono}} = 4 \cdot 1 \cdot 131 = 524 \text{ ps} \rightarrow 1.908 \text{ GHz}$$



$$\text{Speedup} = \frac{1}{0.6 + \frac{0.4}{2}}$$

2. Calcular  $T_{ex}$  y Sup

$$i) T_{ex \text{ Multi}} = 8.205 \cdot 68 \text{ ps} = 111520 \text{ ps}$$

$$111520 - 2720 = 108800 \rightarrow 97.5\%$$

$$a = 100 \quad S_{up} = \frac{1}{0.03 + \frac{0.97}{100}} = 0.039$$

$$i) T_{ex\ multi} = 5 \cdot 202 \cdot 68 = 69680 \text{ ps}$$

$$69680 - 6800 = 68000 \rightarrow 99\%$$

$$a = 100 \quad S_{up} = \frac{1}{0'03 + \frac{0'4}{100}} = 0'01$$

$$iii) T_{ex\ multi} = 4 \cdot 203 \cdot 68 = 55210 \text{ ps}$$

$$55210 - 8'6 = 54394 \rightarrow 98\%$$

$$a = 100 \quad S_{up} = \frac{1}{0'03 + \frac{0'49}{100}} = 0'01$$

$$3. \quad I = 100 \cdot 10^6 \quad GIPS = \frac{100 \cdot 10^6 \cdot 10^6}{6250 \cdot 10^{-3}} = 16 \cdot 10^3$$

$$Flops = 80 \cdot 10^6$$

$$P = 200 \text{ W}$$

$$T_{cpu} = 6250$$

$$GFlop = \frac{80 \cdot 10^6}{6250 \cdot 10^{-3}} = 0'0128$$

$$EPI = \frac{200 \cdot 6250 \cdot 10^{-3}}{100 \cdot 10^6 \cdot 10^6} = 12'5 \cdot 10^{-3} \text{ nJ}$$

- Si son ISA distintas, el numero de instrucciones varia porque GIPS o GFLOPS no son utiles

Para EPI no importa ISA

↳ Arquitectura del conjunto de instrucciones

- Si son las mismas ISA, los GIPS y GFlop se pueden usar como EPI